

Linear regression demonstration

Notes

Synthetic dataset with Gaussian noise, ipsum lorem, ipsum lorem, ipsum lorem, ipsum lorem, ipsum lorem, ipsum lorem, ipsum lorem, ipsum lorem

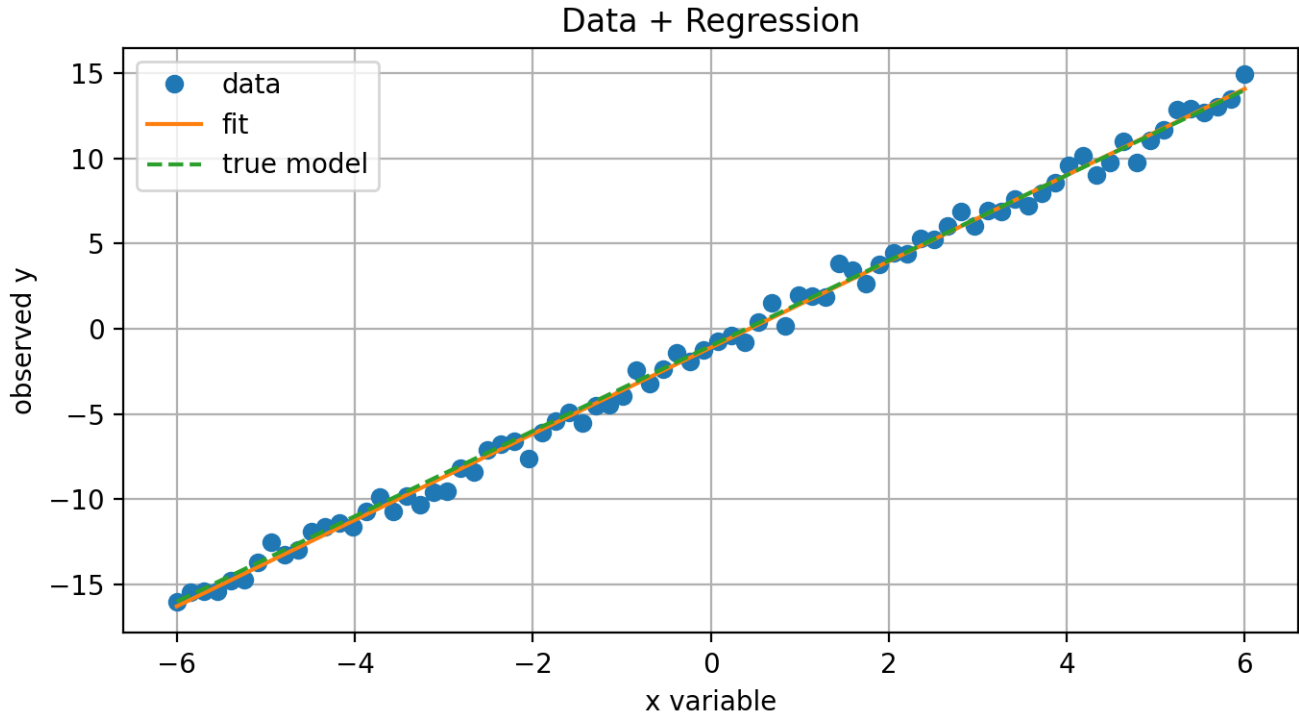


Figure 1: Graph showing linear regression of Demo dataset + true model of Linear regression demonstration

Fit summary

Key	Value
Model	linear
R^2	0.9966
RMSE	0.5212
χ^2	60.37
χ^2_{red}	0.7739
dof	78
weighted	Yes
absolute sigma	Yes
method	None
maxfev	100000
bounds	Yes
initial guess p0	(not stored)

$$f(x) = 2.529x - 1.106$$

Fitted parameters

Parameter	Estimate
A	2.529 ± 0.01912
B	-1.106 ± 0.06708

Dataset info

Key	Value
Name	Demo dataset
n (samples)	80
d (features)	1
x label	x variable
y label	observed y
yerr	Provided
xerr	None
x min/max	x1: [-6, 6]
y min/max	[-16, 14.91]